Hospital infection control: can earlier transfers help predict later clusters?

Choose **this option or the power-network option**, not both. This small, synthetic dataset describes wards and transfers in a regional hospital group. Each ward is a node. Transfers observed **before** the prediction point are directed, weighted links. At that point some wards already have active clusters. Predict a **new** simulated cluster during the next 14 days only among wards without an active cluster.

Work through Part 1 (data and EDA), Part 2 (network), and Part 3 (comparison). Most code is provided. Replace six obvious `___` markers using the hints beside them, then complete the short written responses. Keep code, plots and short answers visible when you submit the notebook. No information from `new_cluster_later` may enter a model feature.

Portfolio summary

- **Author:** marielandreou (GitHub)
- **Course:** Complex Systems and Applications, final project
- **Option chosen:** Option 1, Hospital infection control
- **Data:** `hospital_wards.csv` and `hospital_transfers.csv` (synthetic course data, loaded from `https://raw.githubusercontent.com/mikeauth/leap/main/hospital`)
- **How to run:** *Kernel → Restart & Run All.* It needs internet access, or both CSVs placed beside this notebook.

Question. Among the 102 wards without an active cluster, do directed transfers from the previous 14 days help predict a new simulated cluster in the next 14 days beyond occupancy and device use?

Approach. I built a directed, weighted transfer network (120 wards, 420 links) and computed each ward's total incoming transfer volume and the volume it received from the 18 wards already active. Two logistic regressions, one with ward attributes only and one adding both network features, were trained and tested on the same stratified 70/30 split. Scaling was fitted inside the training pipeline.

Result. Yes, on this test set. F1 rose from 0.435 to 0.737 and ROC AUC from 0.693 to 0.904, and false alarms fell from 6 to 0, but 5 of 12 later clusters were still missed. Across 100 repeated cross-validation folds the network model scored higher in 98%, and two placebo checks showed no gain.

Main limitation. The data are synthetic and small, so this is an association in one sample. It is not evidence that transfers cause infections or that the model is fit for clinical decisions.

Contents. Part 1, data and EDA · Part 2, directed network · Part 3, required model comparison · Part 4, extension (robustness, ablation, placebo controls, error and threshold analysis) · Final interpretation · Reflection

Part 1: load the prepared data and explore it

The two CSVs have already been constructed. The notebook downloads them from their published course URLs; if the address is temporarily unavailable, put both provided CSVs beside this notebook. `occupancy_ratio` and `device_use_share` are known before the follow-up period. The later outcome is blank for wards already affected at the prediction point.

In [1]:

```
DATA_BASE_URL = 'https://raw.githubusercontent.com/mikeauth/leap/main/hospital'
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import networkx as nx
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score, ConfusionMatrixDisplay

SEED = 123

def load_course_csv(filename):
    # The course address is the folder containing the prepared CSVs. For offline work, keep a
    # downloaded CSV in the same folder as this notebook.
    url = f'{DATA_BASE_URL}/{filename}'
    try:
        return pd.read_csv(url)
    except Exception:
        local = Path(filename)
        if local.is_file():
            print(f'Using local copy of {filename}; course URL was unavailable.')
            return pd.read_csv(local)
        raise FileNotFoundError(f'Cannot read {url}. Download {filename} next to the notebook or ask for t
he course URL.')

wards = load_course_csv('hospital_wards.csv')
transfers = load_course_csv('hospital_transfers.csv')
assert wards['ward_id'].is_unique
assert {'ward_id','occupancy_ratio','device_use_share','active_cluster_at_start','new_cluster_later'} <= s
et(wards)
assert {'source_ward_id','destination_ward_id','transfer_count'} <= set(transfers)
display(wards.head())
display(transfers.head())
print('Wards:',len(wards),'Directed links:',len(transfers))
```

	ward_id	occupancy_ratio	device_use_share	active_cluster_at_start	new_cluster_later
0	W001	0.758	0.187	1	NaN
1	W002	0.886	0.338	0	1.0
2	W003	0.533	0.197	0	0.0
3	W004	0.777	0.364	0	1.0
4	W005	0.775	0.446	0	1.0

	source_ward_id	destination_ward_id	transfer_count
0	W001	W006	4
1	W001	W042	6
2	W001	W093	9
3	W001	W111	5
4	W002	W050	3

```
Wards: 120 Directed links: 420
```

Focused EDA: follow the worked code

Exclude wards already affected at the observation point. Count eligible wards and the two later classes. Show one bar chart of the later outcome and two boxplots comparing the original ward attributes by later outcome. In two or three sentences, say whether the groups overlap. Looking at labels for EDA is acceptable; do not use later labels to create predictors.

In [2]:

```
# TODO 1: Fill in the starting-status value for eligible wards.
# Hint: wards['active_cluster_at_start'].eq(0)
eligible_eda = wards.loc[wards['active_cluster_at_start'].eq(0)].copy()
assert eligible_eda['new_cluster_later'].notna().all()
counts = eligible_eda['new_cluster_later'].value_counts().reindex([0.,1.],fill_value=0)
print('Eligible:',len(eligible_eda),'Later clusters:',int(counts.loc[1.]))
fig, axes = plt.subplots(1,3,figsize=(11,3.1))
counts.plot.bar(ax=axes[0],color=['#53799a','#d18166'])
axes[0].set(xlabel='New cluster later (0 or 1)',ylabel='Wards',title='Later outcomes')
for ax,column,label in zip(axes[1:],['occupancy_ratio','device_use_share'],['Occupancy ratio','Device-use
share']):
    groups=[eligible_eda.loc[eligible_eda['new_cluster_later'].eq(k),column] for k in [0,1]]
    ax.boxplot(groups,tick_labels=['No','Yes'])
    ax.set(xlabel='New cluster later',ylabel=label,title=label)
plt.tight_layout();plt.show()
```

```
Eligible: 102 Later clusters: 39
```



Write here: How many wards are eligible? Do their starting attributes separate the later classes cleanly? (Two or three sentences.)

Answer. 102 of the 120 wards are eligible, because 18 already had an active cluster. Of these, 39 later had a new cluster and 63 did not, so about 38% are positive. Wards with a later cluster tend to have higher occupancy (median 0.78 vs 0.66) and somewhat higher device use (median 0.28 vs 0.19). However, the boxes overlap (almost completely for device use), and the full ranges nearly coincide (occupancy 0.49–0.95 for "No" vs 0.56–0.96 for "Yes"), so neither attribute separates the classes cleanly.

Part 2: directed transfer network

A row in `hospital_transfers.csv` means transfers **from** `source_ward_id` **to** `destination_ward_id` in the 14 days before prediction. Use a directed graph and attach `transfer_count` as the link weight. Add all wards first, including isolated wards. Calculate each ward's incoming transfer volume, then the volume received from wards with a cluster **already active** at the prediction point. These use the known initial status, never later labels.

In [3]:

```
G = nx.DiGraph()
G.add_nodes_from(wards['ward_id'])
valid = set(wards['ward_id'])
assert transfers['source_ward_id'].isin(valid).all() and transfers['destination_ward_id'].isin(valid).all(
)
assert (transfers['source_ward_id'] != transfers['destination_ward_id']).all()
assert not transfers.duplicated(['source_ward_id','destination_ward_id']).any()
for row in transfers.itertuples(index=False):
    # TODO 2: Supply the number of transfers as the link weight.
    # Hint: transfer_count=int(row.transfer_count)
    G.add_edge(row.source_ward_id,row.destination_ward_id,transfer_count=int(row.transfer_count))
initial = wards.set_index('ward_id')['active_cluster_at_start']
# TODO 3: Select sources with a cluster already active at the start.
# Hint: initial.loc[src] == 1
wards['incoming_volume'] = [G.in_degree(i,weight='transfer_count') for i in wards['ward_id']]
wards['from_active_volume'] = [sum(d['transfer_count'] for src,_,d in G.in_edges(i,data=True)
                                   if initial.loc[src] == 1) for i in wards['ward_id']]
print('Nodes:',G.number_of_nodes(),'Directed links:',G.number_of_edges())
display(wards[['ward_id','incoming_volume','from_active_volume']].head())
assert (wards['from_active_volume'] <= wards['incoming_volume']).all()
```

```
Nodes: 120 Directed links: 420
```

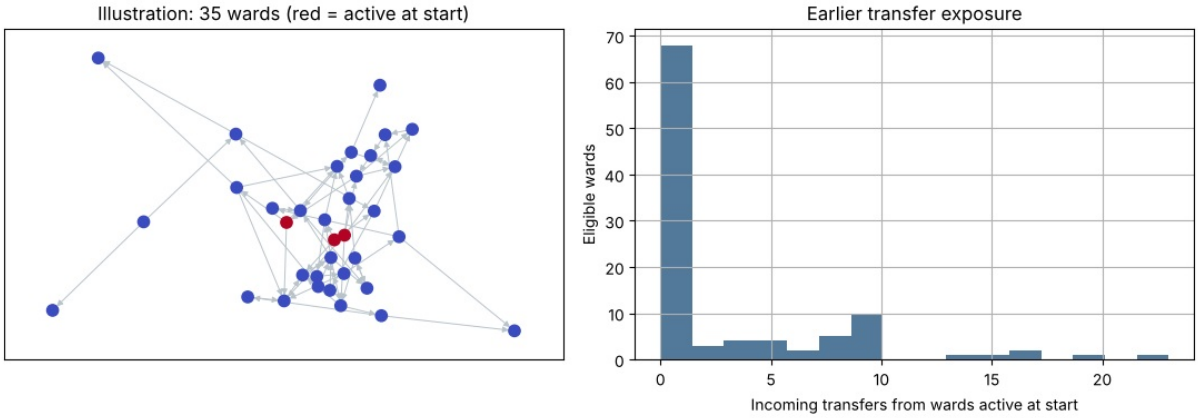
	ward_id	incoming_volume	from_active_volume
0	W001	19	0
1	W002	42	15
2	W003	14	3
3	W004	30	0
4	W005	8	0

Visualise and interpret the network

Make one readable illustration using about 35 wards with the largest incoming volume. Colour nodes by **initial** cluster status; do not colour them by later outcomes. A plotted subset is for illustration only: **use all wards and all links** for feature calculations. Plot a histogram of `from_active_volume` among eligible wards. Write what a high value means and why edge direction matters.

In [4]:

```
shown=wards.nlargest(35,'incoming_volume')['ward_id'].tolist()
small=G.subgraph(shown)
fig,axes=plt.subplots(1,2,figsize=(11,4))
pos=nx.spring_layout(small,seed=42)
nx.draw_networkx(small,pos,ax=axes[0],with_labels=False,node_size=55,
    node_color=[initial.loc[i] for i in small.nodes],cmap='coolwarm',vmin=0,vmax=1,
    edge_color='#b8c3ca',width=.7,arrows=True,arrowsize=8)
axes[0].set_title('Illustration: 35 wards (red = active at start)')
wards.loc[wards['active_cluster_at_start'].eq(0),'from_active_volume'].hist(
    ax=axes[1],bins=16,color='#53799a')
axes[1].set(xlabel='Incoming transfers from wards active at start',ylabel='Eligible wards',title='Earlier
transfer exposure')
plt.tight_layout();plt.show()
```



Write here: What does high `from_active_volume` mean? Why is it different from outgoing transfers? (Two sentences.)

Answer. A high `from_active_volume` means the ward received many patients in the previous 14 days from wards that already had an active cluster at the observation point. In other words, it measures the ward's recorded exposure to the affected part of the network. Among eligible wards it ranges from 0 to 23, and 65 of the 102 received no such transfers. Direction matters because a transfer from ward X to ward Y moves patients *into* Y, so only incoming links describe what arrives at a ward, while outgoing transfers describe what it sends elsewhere. Part 4 tests this: the reversed measure, volume *sent to* active wards, adds almost nothing.

Part 3: compare logistic regression fairly

Only wards without an active cluster are eligible. Model A uses `occupancy_ratio` and `device_use_share`. Model B adds `incoming_volume` and `from_active_volume`. Use the **same wards and the same stratified train/test split**. Fit each scaler inside a pipeline trained only on training rows. Compare accuracy, F1 for later clusters, ROC AUC from probabilities, and both confusion matrices. With a small fixed sample, different scores can move in different directions.

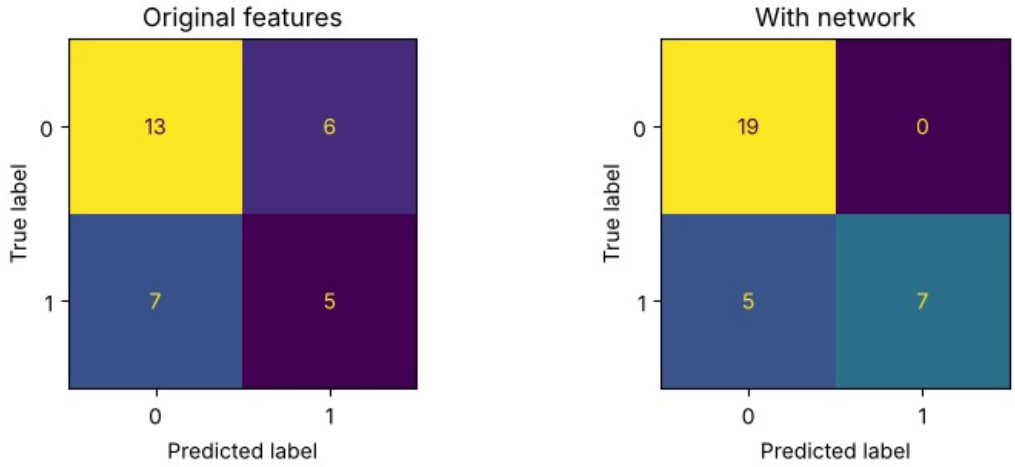
In [5]:

```
eligible = wards.loc[wards['active_cluster_at_start'].eq(0)].set_index('ward_id').copy()
y = eligible['new_cluster_later'].astype(int)
# TODO 4: Set aside 30% of eligible observations for testing.
# Hint: test_size=.30
train_ids,test_ids = train_test_split(y.index,test_size=.30,stratify=y,random_state=SEED)
original=['occupancy_ratio','device_use_share']
# TODO 5: Add the measure of transfers from wards active at the start.
# Hint: with_network = original + ['incoming_volume', 'from_active_volume']
with_network=original+['incoming_volume','from_active_volume']

def fit_and_score(columns):
    model=make_pipeline(StandardScaler(),LogisticRegression(max_iter=1000))
    model.fit(eligible.loc[train_ids,columns],y.loc[train_ids])
    predicted=model.predict(eligible.loc[test_ids,columns])
    probabilities=model.predict_proba(eligible.loc[test_ids,columns])[:,1]
    scores={'Accuracy':accuracy_score(y.loc[test_ids],predicted),
            'F1':f1_score(y.loc[test_ids],predicted,zero_division=0),
            # TODO 6: Use the predicted probability for ROC AUC.
            # Hint: roc_auc_score(y.loc[test_ids], probabilities)
            'ROC AUC':roc_auc_score(y.loc[test_ids],probabilities)}
    return scores,predicted

base_scores,base_pred=fit_and_score(original)
network_scores,network_pred=fit_and_score(with_network)
results=pd.DataFrame([base_scores,network_scores],index=['Original ward features','Ward + network features
'])
display(results.round(3))
fig,axes=plt.subplots(1,2,figsize=(8,3.3))
for ax,pred,title in zip(axes,[base_pred,network_pred],['Original features','With network']):
    ConfusionMatrixDisplay.from_predictions(y.loc[test_ids],pred,ax=ax,labels=[0,1],colorbar=False)
    ax.set_title(title)
plt.tight_layout();plt.show()
```

	Accuracy	F1	ROC AUC
Original ward features	0.581	0.435	0.693
Ward + network features	0.839	0.737	0.904



Part 4 (extension): is the gain robust, and where does it come from?

The required comparison in Part 3 is the headline answer, and nothing below replaces it. The test set has only 31 wards, 12 of which later had a cluster, so a single ward changing class moves F1 by about 0.06. To check that the result is not an accident of one split, I added four checks. As in Part 3, `new_cluster_later` is used only as the target and never to build a feature.

1. **Repeated cross-validation.** Stratified 5-fold CV repeated 20 times gives 100 paired train/test evaluations on the same 102 wards. The scaler and model are refitted inside every training fold.
2. **Ablation.** Each network feature is added on its own to see which one carries the signal.
3. **Two placebo controls.**
   - *Direction reversed:* the volume a ward **sent to** active wards, computed from the same graph and the same initial status. If edge direction did not matter, this should help as much as `from_active_volume`.
   - *Shuffled:* the two network features are randomly permuted across wards. This keeps their distribution but breaks the link to each ward. Any gain here would mean the improvement came from simply adding columns.
4. **Error and threshold analysis** of the clusters the network model missed (4b and 4c).

In [6]:

```
# Part 4a: repeated cross-validation, ablation and placebo controls.
from sklearn.base import clone
from sklearn.model_selection import RepeatedStratifiedKFold, StratifiedKFold, cross_val_predict

pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
network_cols = ['incoming_volume', 'from_active_volume']

# Direction placebo: transfers SENT TO wards active at the start (out-edges). It uses only the
# earlier graph and the initial status, never later labels.
eligible['to_active_volume'] = pd.Series(
    {i: sum(d['transfer_count'] for _, dst, d in G.out_edges(i, data=True) if initial.loc[dst] == 1)
     for i in eligible.index})

feature_sets = {
    'A: ward attributes only': original,
    'A + incoming volume': original + ['incoming_volume'],
    'A + volume from active wards': original + ['from_active_volume'],
    'B: A + both network features': with_network,
    'Placebo: A + volume sent to active wards': original + ['to_active_volume'],
    'Placebo: A + shuffled network features': with_network,  # columns permuted in each fold
}
cv = RepeatedStratifiedKFold(n_splits=5, n_repeats=20, random_state=SEED)
rng = np.random.default_rng(SEED)
rows = []
for fold, (tr, te) in enumerate(cv.split(eligible, y)):
    shuffled = eligible.copy()
    shuffled[network_cols] = shuffled[network_cols].to_numpy()[rng.permutation(len(shuffled))]
    for name, cols in feature_sets.items():
        X = shuffled if 'shuffled' in name else eligible
        fitted = clone(pipe).fit(X.iloc[tr][cols], y.iloc[tr])
```

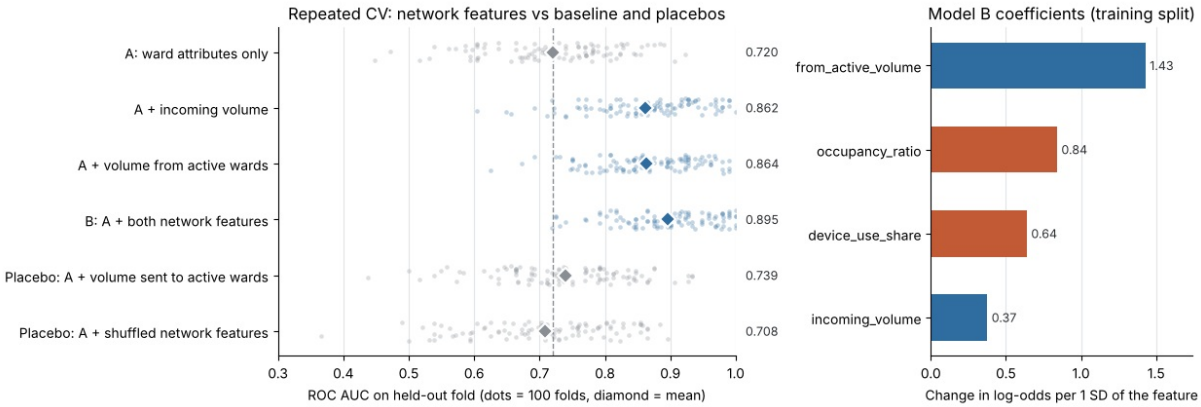
```
        prob = fitted.predict_proba(X.iloc[te][cols])[:, 1]
        rows.append({'fold': fold, 'model': name,
                     'ROC AUC': roc_auc_score(y.iloc[te], prob),
                     'F1': f1_score(y.iloc[te], (prob >= .5).astype(int), zero_division=0)})
cv_results = pd.DataFrame(rows)

base_name = 'A: ward attributes only'
wide = cv_results.pivot(index='fold', columns='model', values='ROC AUC')
cv_summary = cv_results.groupby('model', sort=False).agg(
    **{'Mean ROC AUC': ('ROC AUC', 'mean'), 'SD ROC AUC': ('ROC AUC', 'std'), 'Mean F1': ('F1', 'mean')})
cv_summary['Mean AUC gain vs A'] = wide.sub(wide[base_name], axis=0).mean()
cv_summary['Share of folds beating A'] = wide.gt(wide[base_name], axis=0).mean()
cv_summary.loc[base_name, ['Mean AUC gain vs A', 'Share of folds beating A']] = np.nan
display(cv_summary.round(3))

# Refit the two Part 3 models on the same training rows (identical to fit_and_score) so they can be inspec
ted.
model_A = clone(pipe).fit(eligible.loc[train_ids, original], y.loc[train_ids])
model_B = clone(pipe).fit(eligible.loc[train_ids, with_network], y.loc[train_ids])
coefs = pd.Series(model_B[-1].coef_[0], index=with_network)

BLUE, ORANGE, GREY, INK = '#2f6da0', '#c25a35', '#8a8f96', '#2b2f33'
fig, axes = plt.subplots(1, 2, figsize=(12.5, 4.4), gridspec_kw={'width_ratios': [1.75, 1]})
ax = axes[0]
names = list(feature_sets)[::-1]
for k, name in enumerate(names):
    vals = cv_results.loc[cv_results['model'].eq(name), 'ROC AUC'].to_numpy()
    colour = GREY if (name == base_name or name.startswith('Placebo')) else BLUE
    jitter = np.random.default_rng(k).uniform(-.17, .17, len(vals))
    ax.scatter(vals, k + jitter, s=11, alpha=.28, color=colour, linewidths=0)
    ax.plot(vals.mean(), k, marker='D', ms=8, color=colour, mec='white', mew=1.5, zorder=3)
    ax.text(1.015, k, f'{vals.mean():.3f}', va='center', fontsize=9, color=INK)
ax.axvline(wide[base_name].mean(), color=GREY, ls='--', lw=1)
ax.set_yticks(range(len(names)), names)
ax.set_xlim(.3, 1.0)
ax.set(xlabel='ROC AUC on held-out fold (dots = 100 folds, diamond = mean)',
       title='Repeated CV: network features vs baseline and placebos')
ax = axes[1]
order = coefs.sort_values().index
ax.barh(range(len(order)), coefs[order], height=.55,
        color=[BLUE if c in network_cols else ORANGE for c in order])
for k, c in enumerate(order):
    ax.text(coefs[c] + .03, k, f'{coefs[c]:.2f}', va='center', fontsize=9, color=INK)
ax.set_yticks(range(len(order)), order)
ax.set_xlim(0, coefs.max() * 1.22)
ax.set(xlabel='Change in log-odds per 1 SD of the feature', title='Model B coefficients (training split)')
for a in axes:
    a.spines[['top', 'right']].set_visible(False)
    a.grid(axis='x', color='#e3e6e9', lw=.8)
    a.set_axisbelow(True)
plt.tight_layout(); plt.show()
```

model	Mean ROC AUC	SD ROC AUC	Mean F1	Mean AUC gain vs A	Share of folds beating A
A: ward attributes only	0.720	0.099	0.533	NaN	NaN
A + incoming volume	0.862	0.084	0.730	0.142	0.96
A + volume from active wards	0.864	0.077	0.698	0.144	1.00
B: A + both network features	0.895	0.072	0.722	0.175	0.98
Placebo: A + volume sent to active wards	0.739	0.100	0.536	0.019	0.59
Placebo: A + shuffled network features	0.708	0.105	0.512	-0.013	0.34



What 4a shows.

- **The gain is robust.** Model B averages ROC AUC 0.895 against 0.720 for Model A, a gain of +0.175, and it beats A in 98 of 100 paired folds. Its F1 averages 0.722 against 0.533. The single-split gain in Part 3 (+0.21 AUC) is slightly larger than average, which is expected from a 31-ward test set, but it points the same way.
- **Both features help, and they overlap.** Each one alone adds about +0.14 AUC, and together they add +0.175. The features are correlated (r ≈ 0.59 among eligible wards), because a ward that receives many transfers is also more likely to receive some from active wards. In Model B, `from_active_volume` has by far the largest coefficient (1.43 vs 0.37 for `incoming_volume`).
- **Direction matters.** Reversing the edges (volume *sent to* active wards) adds only +0.02 AUC and beats A in 59% of folds, about the rate of a coin flip. What predicts a cluster is what a ward receives, not what it sends.
- **The gain is not an artefact of adding columns.** With the network features shuffled across wards, AUC falls slightly below A (−0.01, beating A in only 34% of folds).

These are associations in synthetic data. The construction note says the generator was built so that incoming volume and exposure to active wards affect the outcome. The checks therefore show that the models recover a signal that was placed in the data, which is a useful sanity test, not a discovery about real hospitals.

4b: which clusters does the network model miss?

The final interpretation asks about errors. Below are the 12 test wards that later had a cluster, sorted by Model B's predicted probability, along with both models' probabilities (≥ 0.5 counts as flagged).

In [7]:

```
test_view = eligible.loc[test_ids, with_network].copy()
test_view['p_A'] = model_A.predict_proba(eligible.loc[test_ids, original])[:, 1]
test_view['p_B'] = model_B.predict_proba(eligible.loc[test_ids, with_network])[:, 1]
later = test_view[y.loc[test_ids].eq(1)].sort_values('p_B').copy()
later['Model A'] = np.where(later['p_A'] >= .5, 'caught', 'missed')
later['Model B'] = np.where(later['p_B'] >= .5, 'caught', 'missed')
display(later.round(2))
print('Median from_active_volume of clusters Model B missed:',
      later.loc[later['Model B'].eq('missed'), 'from_active_volume'].median(),
      '| caught:', later.loc[later['Model B'].eq('caught'), 'from_active_volume'].median())
```

ward_id	occupancy_ratio	device_use_share	incoming_volume	from_active_volume	p_A	p_B	Model A	Model B
W030	0.75	0.11	25	0	0.23	0.15	missed	missed
W071	0.58	0.14	43	2	0.13	0.18	missed	missed
W099	0.72	0.33	28	0	0.46	0.33	missed	missed
W070	0.70	0.24	12	5	0.32	0.38	missed	missed
W004	0.78	0.36	30	0	0.58	0.48	caught	missed
W011	0.95	0.41	13	0	0.81	0.65	caught	caught
W009	0.72	0.31	26	6	0.44	0.68	missed	caught
W054	0.95	0.32	47	0	0.71	0.78	caught	caught
W110	0.83	0.22	12	9	0.46	0.79	missed	caught
W012	0.86	0.21	21	9	0.47	0.85	missed	caught
W039	0.83	0.44	23	8	0.73	0.93	caught	caught
W065	0.93	0.22	34	14	0.57	0.98	caught	caught

```
Median from_active_volume of clusters Model B missed: 0.0 | caught: 8.0
```

What 4b shows. The five clusters Model B missed were all in wards with little or no recorded exposure: three received no transfers from active wards and the other two received 2 and 5. The clusters it caught had a median exposure of 8. Two caught wards had zero exposure but very high occupancy (0.95) or very high incoming volume (47). One miss, W004, had actually been caught by Model A (0.58) but fell just below the threshold in Model B (0.48), because its lack of exposure lowered its score. So the network model is better on average, not on every ward. The model therefore mostly finds clusters along the transfer paths from affected wards. A cluster that arises without that recorded exposure looks like an ordinary ward to it. This is a systematic blind spot rather than random noise, and it matters for the ethics question: wards outside the main transfer routes would be under-flagged.

4c: trading missed clusters against false alarms

At the default threshold of 0.5, Model B raised no false alarms but missed 5 of 12 clusters. In infection control a miss is usually costlier than a false alarm, but where to set the threshold is a policy choice. To show the trade-off without tuning on the test set, I chose the threshold using **training wards only**: 5-fold out-of-fold predictions on the 71 training wards, keeping the threshold with the highest out-of-fold F1. I then applied it to the test set once.

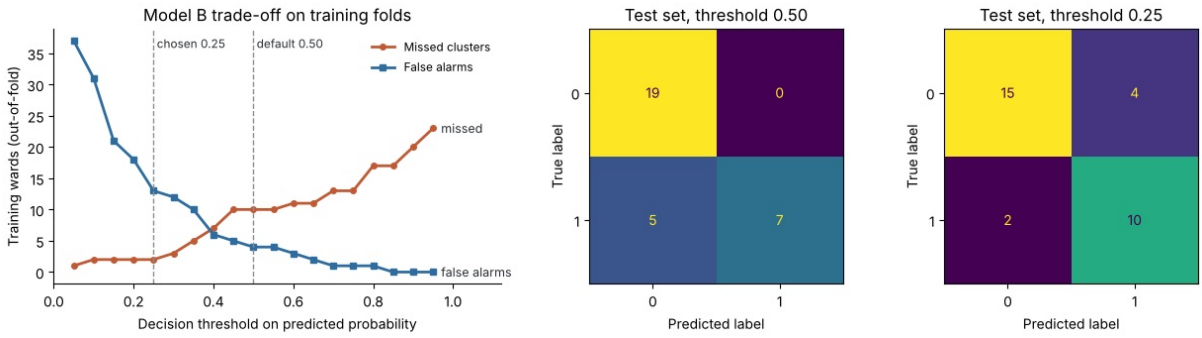
In [8]:

```
y_tr = y.loc[train_ids]
oof = cross_val_predict(clone(pipe), eligible.loc[train_ids, with_network], y_tr,
                        cv=StratifiedKFold(5, shuffle=True, random_state=SEED), method='predict_proba')[:,
1]
thresholds = np.round(np.arange(.05, .96, .05), 2)
missed = [int(((oof < t) & (y_tr == 1)).sum()) for t in thresholds]
false_alarms = [int(((oof >= t) & (y_tr == 0)).sum()) for t in thresholds]
oof_f1 = [f1_score(y_tr, (oof >= t).astype(int), zero_division=0) for t in thresholds]
chosen = float(thresholds[int(np.argmax(oof_f1))])
print(f'Threshold chosen on training folds only: {chosen:.2f}  (out-of-fold F1 {max(oof_f1):.3f})')

p_test = test_view['p_B']
y_te = y.loc[test_ids]
for label, t in [('default 0.50', .5), (f'chosen {chosen:.2f}', chosen)]:
    pred = (p_test >= t).astype(int)
    print(f'Test set, threshold {label}: F1 {f1_score(y_te, pred):.3f}, '
          f'missed {int(((pred == 0) & (y_te == 1)).sum())} of {int(y_te.sum())}, '
          f'false alarms {int(((pred == 1) & (y_te == 0)).sum())} of {int((y_te == 0).sum())}')

fig, axes = plt.subplots(1, 3, figsize=(12.5, 3.6), gridspec_kw={'width_ratios': [1.6, 1, 1]})
ax = axes[0]
ax.plot(thresholds, missed, color=ORANGE, lw=2, marker='o', ms=4, label='Missed clusters')
ax.plot(thresholds, false_alarms, color=BLUE, lw=2, marker='s', ms=4, label='False alarms')
ax.text(thresholds[-1] + .02, missed[-1], 'missed', va='center', fontsize=9, color=INK)
ax.text(thresholds[-1] + .02, false_alarms[-1], 'false alarms', va='center', fontsize=9, color=INK)
for t, txt in [(.5, 'default'), (chosen, 'chosen')]:
    ax.axvline(t, color=GREY, ls='--', lw=1)
    ax.text(t, ax.get_ylim()[1] * .97, f' {txt} {t:.2f}', fontsize=8.5, color=INK, va='top')
ax.set_xlim(0, 1.12)
ax.set(xlabel='Decision threshold on predicted probability',
       ylabel='Training wards (out-of-fold)', title='Model B trade-off on training folds')
ax.legend(frameon=False, loc='upper right', fontsize=8.5)
ax.spines[['top', 'right']].set_visible(False)
for ax, t in zip(axes[1:], [.5, chosen]):
    ConfusionMatrixDisplay.from_predictions(y_te, (p_test >= t).astype(int), ax=ax, labels=[0, 1], colorba
r=False)
    ax.set_title(f'Test set, threshold {t:.2f}')
plt.tight_layout(); plt.show()
```

```
Threshold chosen on training folds only: 0.25  (out-of-fold F1 0.769)
Test set, threshold default 0.50: F1 0.737, missed 5 of 12, false alarms 0 of 19
Test set, threshold chosen 0.25: F1 0.769, missed 2 of 12, false alarms 4 of 19
```



What 4c shows. The threshold chosen on the training folds was 0.25. On the test set it would have reduced missed clusters from 5 to 2 at the cost of 4 false alarms, and F1 would have risen from 0.737 to 0.769. This does not make a "better model". It is the same model with a different balance of errors. Which balance is acceptable depends on the cost of an unnecessary screening round compared with an undetected cluster. That decision belongs to the infection-prevention team, not to the metric.

Final interpretation and portfolio summary

Prompt: 100–180 words. Answer the question, compare F1 and ROC AUC, describe an error, distinguish the two network features, and consider the ethics.

Did adding earlier directed transfers help predict new clusters on this test set? Yes. On the same 31 test wards, F1 rose from 0.435 to 0.737, ROC AUC from 0.693 to 0.904, and accuracy from 0.581 to 0.839. All three metrics agree, so the result is not mixed. Model A raised 6 false alarms and missed 7 of 12 clusters. Model B raised none but still missed 5, mostly wards with no recorded exposure. Total incoming volume measures how many patients a ward receives from anywhere. Volume from initially affected wards counts only patients arriving from wards that already had a cluster, and it carried the larger coefficient. If predictions guided infection-control attention, a false alarm could divert screening, isolation beds and staff time and worry patients. A missed cluster could leave patients and staff unprotected. The infection-prevention team and ward clinicians should review flags before acting and audit misses across ward types. Synthetic data cannot show that transfers cause infections.

Reflection

What worked well. The pipeline design made the comparison fair by construction: both models share one split, and scaling is learned only from training rows. The placebo controls were the most useful addition. They turned "the score went up" into "the gain depends on *who sent* the patients, not on adding more columns".

What was difficult. With only 31 test wards, one ward changing class moves F1 by about 0.06, so a single split can overstate or understate a difference. Repeated cross-validation was needed to judge the size of the gain. Timing also needed constant care: every feature, including the placebo, had to use only the initial status and the earlier transfers, never later labels.

What I would improve with more time.

- Use a time-ordered validation (train on one period, test on the next) instead of random splits.
- Try exposure measures that follow longer transfer chains, such as two-step paths or a PageRank-style score.
- Calibrate the predicted probabilities before using them to rank wards.
- For any real use, validate on independent hospital data together with clinicians.